Sexual Hegemons and Involuntary Celibates

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Introduction

Background

- **Less intimate partnership:** (Ueda et al. 2020; Twenge, Sherman, and Wells 2017; Lei and South 2021)
  - **Fewer adults in steady relationships**.
  - **Less sex:** U.S. men 18–44 with no sex in the past year — 9.5% $\rightarrow$ 16.5%, 2000–02 to 2016–18.

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- **Cheaper meeting:** (Rosenfeld and Thomas 2012; Rosenfeld, Thomas, and Hausen 2019)
  - **Most dating starts with apps.**

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- **Cheaper meeting:** (Rosenfeld and Thomas 2012; Rosenfeld, Thomas, and Hausen 2019)
  - **Most dating starts with apps.**
- **More inequality among men:** (Goldgeier 2017; Bruch and Newman 2018)
  - **Hinge profile likes:** Gini 0.54 for men, 0.38 for women.

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Q2. **Hierarchy or organization?** Does the hierarchy of desirability itself create **sexual hegemons** and **involuntary celibates**, or is this a result of the way the market is organized?

- **Hegemons:** men who repeatedly match more than average.
- **Celibates:** men rejected by every woman who learns his type, only matching by mistake.

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Q3. **Efficiency and distribution.**

(a) **Efficiency:** Could some be made better off with no one made worse off? If so, why does the market fail to make the change?

(b) **Distribution:** Judged by how unequal men's welfare is, how far is the market from the best assignment?

What I Do

- **A Diamond–Mortensen–Pissarides search model**, extended to **two-sided matching** in dating. Men and women differ in productivity and desirability; men want their own kind, women share one ranking. No transfers.

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- **Contact and screening costs** (Bagga and Mann 2026), on both sides: opening a conversation, and learning what a profile hides, with error.
- **Endogenous separations**: each partner's taste for the match is redrawn now and then; either leaves when searching again is worth more (Mortensen and Pissarides 1994).

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Findings

1. **Break-ups alone make outcomes unequal among men.** With every type visible and contact free, every man matches at the same rate, but women leave the men they value less sooner: the most desirable are partnered longest, the least desirable change partners most. Costly contact adds unequal match rates. Props. 3.1′, 3.4′, 3.5′

1: the two measures of concentration rank types oppositely when contact is free. 2: where break-ups respond strongly to match rates. 3: exact screening. 4: at a given pool, where men accept the sub-market's women alike; with fast redraws men's response can reverse it.

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4. **Cheaper contact cuts both ways.** Where types are visible, every man matches more often; where they are hidden, women get more options and raise their standards at a given pool. Props. 3.4′, 3.6(iii)′

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Literature

Literature (1/2): Search and Matching Foundations

- **Equilibrium search and matching**: (Diamond 1982; Mortensen and Pissarides 1994; Pissarides 2000; Petrongolo and Pissarides 2001); used for marriage markets (Mortensen 1988; Burdett and Coles 1997; Shimer and Smith 2000); used for dating markets (Kanoria and Saban 2021; Antler and Bachi 2022)
- **Directed search**: (Moen 1997; Menzio 2007; Jacquet and Tan 2007; Wright et al. 2021).
- **Search externalities (no posting price)**: competitive search is efficient when a posting side sets a price (Hosios 1990; Shi 2001; Eeckhout and Kircher 2010); two-sided dating has none (Goody and Tambiah 1973; Borgerhoff Mulder 1988).
- **Sorting without transfers**: from mutual acceptance and congestion (Gale and Shapley 1962; Bloch and Ryder 2000; Adachi 2003; Smith 2006; Lauermann and Nöldeke 2014; Chade, Eeckhout, and Smith 2017).

Literature (2/2): What This Adds

Technical addition: two-sided DMP matching in Moen (1997) sub-markets that **both** sides choose, with no posting and no transfers, under **two-sided contact and screening** (Bagga and Mann 2026), with **break-ups each partner chooses** (Mortensen and Pissarides 1994).

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  - **Here:** if signals are poor, **exit** degrades pool quality — several equilibria.
- **Misspecified beliefs** (Fudenberg and Levine 1993; Esponda and Pouzo 2016; Jehiel 2018).
  - **Here:** one is more likely to accept desirable candidates from a pool $\Rightarrow$ biased beliefs as to a pool's composition.

The Model

Agents and Types

- Two sexes, men ♂ and women ♀.
- Each agent has a fixed type:

$$z = (\omega, \tau), \qquad \omega = \text{productivity}, \quad \tau = \text{desirability}, \tag{1}$$

  each high (H) or low (L).
- **4 types per sex:** $z_{H,H}$, $z_{H,L}$, $z_{L,H}$, $z_{L,L}$.

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- **Baseline:** $N^{♂} = N^{♀}$ and $\phi^{♂}(z) = \phi^{♀}(z)$ for every z.

Type-conditional Male Preferences for Females

Matched male flow utility $u(z^{♀}; z^{♂}) = f(z^{♀}; z^{♂}) + b(z^{♂})$ with

$$f(z^{♀}; z^{♂}) = \underbrace{f_0}_{\text{match premium}} - \underbrace{\left(d_\omega \mathbf{1}\{\omega^{♀} \neq \omega^{♂}\} + d_\tau \mathbf{1}\{\tau^{♀} \neq \tau^{♂}\}\right)}_{\text{mismatch penalties}} \tag{2}$$

Unmatched $u(\varnothing; z^{♂}) = b(z^{♂})$.

- **Men most prefer a woman of the same type ranking.**

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- **Men most prefer a woman of the same type ranking.**
- Likes attract (Buston and Emlen 2003).
- Men tend to not value human capital beyond their own (Fisman et al. 2006; Bertrand, Kamenica, and Pan 2015; Neyt, Vandenbulcke, and Baert 2019).
  - Some caveats ... (Todd et al. 2007; Binder and Lam 2022; Egebark et al. 2021; Hederos and Stenberg 2022).

Monotonic Female Preferences for Males

Matched female flow utility $u(z^{\male}; z^{\female}) = g(z^{\male}) + b(z^{\female})$ with

$$g(z^{\male}) = \alpha_\omega(\omega^{\male}) + \alpha_\tau(\tau^{\male}) \tag{3}$$

with $\Delta\alpha_\omega > \Delta\alpha_\tau > 0$ and $\Delta\alpha_j \equiv \alpha_j(H) - \alpha_j(L), \ \forall j \in \{\omega, \tau\}$

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- Women weight productivity more heavily than desirability.
- Women weight resources more than men do (Buss 1989; Walter et al. 2020; Li et al. 2002).
- Women of every income favor higher-income men (Ong and Wang 2015; Hitsch, Hortaçsu, and Ariely 2010b; Hitsch, Hortaçsu, and Ariely 2010a).
- Women tend to be choosier (Todd et al. 2007; Kreager et al. 2014).
  - Some caveats ... (Eastwick and Finkel 2008; Eastwick et al. 2014; Egebark et al. 2021; Belot and Francesconi 2013).

What Monotonicity & $\Delta\alpha_\omega > \Delta\alpha_\tau$ Assumptions Do

- Monotonicity ($\Delta\alpha_j > 0$) ranks the end points — $z_{H,H} \succ z_{L,L}$.
- $\Delta\alpha_\omega > \Delta\alpha_\tau$ ranks the mixed types, productive over desirable — $g(z_{H,L}) - g(z_{L,H}) = \Delta\alpha_\omega - \Delta\alpha_\tau > 0$:

$$g(z_{L,L}) < g(z_{L,H}) < g(z_{H,L}) < g(z_{H,H}). \tag{4}$$

- Reverse it, and ω and τ just trade places.

Time and Match States

- Continuous time, discount rate β.
- Each agent is **matched** or **unmatched**; let $\varnothing$ indicate no match.
- Flow utility adds y^x, each partner's own **match-specific valuation**:

$$u(z^{\female}_{\male}; z^{\male} \mid y^{\male}) = f(z^{\female}_{\male}; z^{\male}) + b(z^{\male}) + y^{\male}, \qquad u(\varnothing; z^x) = b(z^x), \quad \forall x \in \{\male, \female\} \tag{5}$$

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$v^{\male} \equiv f(z^{\female}_{\male}; z^{\male})$, $v^{\female}_{\male} \equiv g(z^{\male})$: what x gets from the partner's **type**; y^x: from **this match**, and it changes (next).

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- **Only the unmatched search**; the matched decide only whether to stay. Unmatched mass of type z and of sex x:

$$n^x(z) \leq N^x \phi^x(z), \qquad n^x = \sum_z n^x(z) \tag{7}$$

How Matches End (1/2): Each Partner's Valuation Is Redrawn

- A new match starts at $y^{\male} = y^{\female} = 0$.

A redraw replaces y^x and does not depend on it. y^x is private: to x, only whether the partner stays matters. The job-destruction margin of Mortensen and Pissarides (1994), with each partner deciding for itself.

How Matches End (1/2): Each Partner's Valuation Is Redrawn

- A new match starts at $y^{\male} = y^{\female} = 0$.
- At Poisson rate θ, **each** partner's valuation is redrawn, independently of the other's:

$$y^x \longrightarrow y' \sim \Upsilon, \qquad \text{a continuous cdf; } \Upsilon = \text{Normal}(0, \sigma_y^2) \text{ in computations} \tag{8}$$

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- **Exit is unilateral:** x ends the match iff y^x falls below a **reservation shock** $\underline{y}^x(z^{\male}, z^{\female}_{\mp})$, where staying and leaving are worth the same — an **equilibrium object**, derived below.

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- Matches also dissolve **exogenously**, at rate $\delta_0 \geq 0$: separations are **semi-endogenous**. Either way, both return to search.

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How Matches End (2/2): The Separation Hazard

▷ Steady State

A match ends at a δ_0 shock, or at a redraw below the redrawer's reservation:

$$\delta(z^{\male}, z^{\female}_{\mp}) = \underbrace{\delta_0}_{\text{exogenous}} + \underbrace{\theta\,\Upsilon\big(\underline{y}^{\male}\big)}_{\text{he redraws, leaves}} + \underbrace{\theta\,\Upsilon\big(\underline{y}^{\female}_{\mp}\big)}_{\text{she redraws, leaves}} \tag{9}$$

Scope: what each partner knows of the other's type is fixed during the match (types visible, or screened exactly; under noisy screening, beliefs held at formation). With Υ symmetric about 0, $\delta < \delta_0 + \theta$ for accepted pairs.

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The hazard is **constant over a match's life**: durations are exponential, with mean $1/\delta(z^{\male}, z^{\female})$.

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- **Redraws off**, $\theta = 0$: $\delta(z^{\male}, z^{\female}_{\mp}) = \delta_0$, the model with one exogenous dissolution rate.
- No accepted match ends at once: both start at 0, above both reservations.

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Why Valuations That Change?

- **Separations that respond to options**: match-specific shocks and reservation rules are how search models make job destruction endogenous (Mortensen and Pissarides 1994; Pissarides 2000); partner search was cast in the same terms early on (Mortensen 1988).
- **Each partner's valuation drifts**: adultery is the most cited cause of dissolution across 160 societies (Betzig 1989), a change on one side.
- **Do platforms raise separations?** Here cheaper contact raises every pair's hazard, through faster matching. Samples of couples formed online do not show faster dissolution (Cacioppo et al. 2013; Potârcă 2020; Rosenfeld, Thomas, and Hausen 2019); they compare meeting channels, not contact costs within one market.

Turnover is now an outcome: what speeds matching also speeds break-ups.

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- **Observable set** $\mathcal{O} \subseteq \{\omega, \tau\}$: the traits seen before meeting
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  - Both traits, one, or neither may be visible.
  - Non-visible traits are hidden.
- **Sub-market** S: indexed by the **visible types** of the men and the women searching in it,

$$S = (z_{\mathcal{O}}^{\male}, z_{\mathcal{O}}^{\female}) \tag{10}$$

Directed Search ▷ 4 Regimes

- **Choosing a sub-market is choosing a type**, up to what $\mathcal{O}$ reveals:
  - A man picks the women's visible type $z^{\female}_{\mathcal{O}}$ (his own $z^{\male}_{\mathcal{O}}$ is fixed).
  - A woman picks the men's, $z^{\male}_{\mathcal{O}}$.

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- **Directed across sub-markets, random within**: you choose the sub-market, never whom you meet in it.
- Composition of each sub-market and the mix of hidden types met there is an **equilibrium outcome** (more later).

Meeting (Matching) Function

- In sub-market S, unmatched men and women meet at the flow (Diamond 1982; Mortensen and Pissarides 1994; Pissarides 2000)

$$M(S) = \xi \, n^{\male}(S)^{1-\psi} \, n^{\female}(S)^{\psi} \tag{11}$$

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- $n^{x}(S)$: unmatched mass of sex x in S.
- Common to every S: meeting efficiency $\xi > 0$ and $\psi \in (0, 1)$, the elasticity of meetings with respect to $n^{\female}(S)$.

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- In sub-market S, unmatched men and women meet at the flow (Diamond 1982; Mortensen and Pissarides 1994; Pissarides 2000)

$$M(S) = \xi \, n^{\male}(S)^{1-\psi} \, n^{\female}(S)^{\psi} \tag{11}$$

- $n^{x}(S)$: unmatched mass of sex x in S.
- Common to every S: meeting efficiency $\xi > 0$ and $\psi \in (0, 1)$, the elasticity of meetings with respect to $n^{\female}(S)$.
- **Tightness**:

$$\Theta(S) = \frac{n^{\male}(S)}{n^{\female}(S)} \tag{12}$$

Meeting Rates, λ^x

- Tightness sets each searcher's **meeting rate**, λ^x.

$$\lambda^{\male}(\Theta(S)) = \frac{M(S)}{n^{\male}(S)} = \xi\,\Theta(S)^{-\psi} \tag{13}$$

$$\lambda^{\female}(\Theta(S)) = \frac{M(S)}{n^{\female}(S)} = \xi\,\Theta(S)^{1-\psi} \tag{14}$$

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- More men — each man meets less often, each woman more often.
- Balanced sub-market, $\Theta(S) = 1$ — $\lambda^x(1) = \xi$ and ψ matters only where tightness differs from 1.
- **Whom you meet** is random: a draw from the unmatched of the opposite sex x in S. With $n^x(z^x|S)$ the unmatched mass in S of type z^x, the meeting probability is

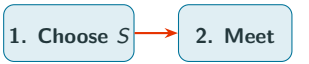
$$\pi(z^x|S) = \frac{n^x(z^x|S)}{n^x(S)} \tag{15}$$

The Path of a Meeting: Up to Screening

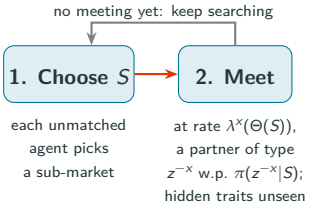
1. Choose S

each unmatched
agent picks
a sub-market

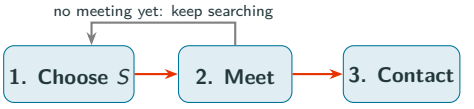
The Path of a Meeting: Up to Screening



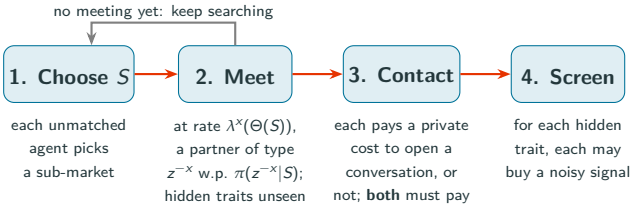
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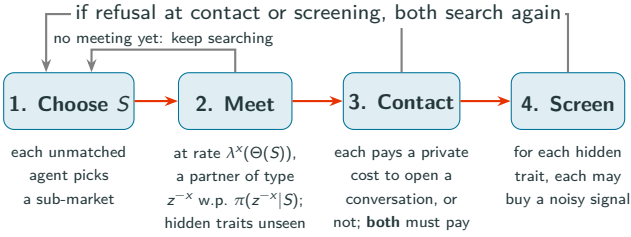
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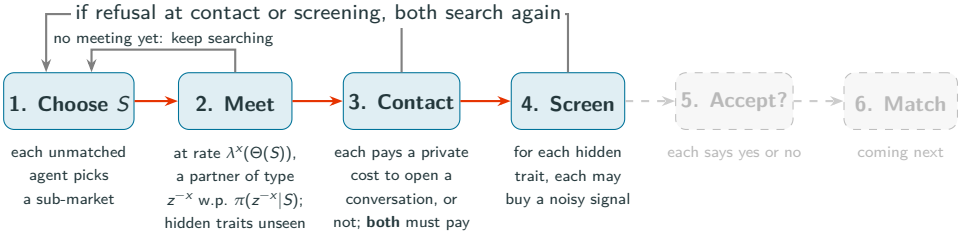
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- $\overline{c}^x(z^x, S)$ is an **equilibrium object**: the most x would pay to talk, its expected gain in continuation value from paying the cost and moving forward vs. walking away.

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- On an app: photos show τ; the chat or first date reveals job and income, ω, with error (some apps, e.g. Raya, opposite).

Screening: Both Sides Screen

	Men screen **defensively**	**Women** screen **acquisitively**
"Favorable" on j	she matches his type on j	he is high on j
Prior q_j	share of matching women among the unmatched women of S	share of high men among the unmatched men of S
Stake: gain if favorable	$d_j/(\beta+\delta_0)$ with redraws off	$\Delta\alpha_j/(\beta+\delta_0)$ with redraws off
Screens ($\Psi_j^x > 0$) only if	he would accept a woman who matches him on j and reject one who does not	she would accept the man high on j and reject the man low on j
Ease of screening	$\gamma_{s,j}^{\male}$	$\gamma_{s,j}^{\female}$

The elasticity η_s is common to both sexes. With redraws the stake is the gap in surplus at formation (Values), and a favorable partner is also likelier to stay. Stakes as shown when the other dimension is visible or independent of j in the pool. A match must survive both sides' screens on every hidden j.

Turning Contact, Screening and Redraws Off

- After a meeting, the two stages multiply the chance that it becomes a match by

$$\underbrace{P^{\male}(z^{\male}, S)\, P^{\female}(z^{\female}, S)}_{\text{both commit to contact}} \times \underbrace{\prod_{j \notin \mathcal{O}} P^{\male}_{s,j}(z^{\male}, z^{\female}, S)\, P^{\female}_{s,j}(z^{\female}, z^{\male}, S)}_{\text{both survive the screens}} \tag{20}$$

- **Contact cost off**: $\gamma^x \to \infty$. The cost support $[0, (\gamma^x)^{-1/\eta}]$ shrinks to $\{0\}$: contact is free, and $P^x(z^x, S) = 1$ wherever the threshold is positive.
- **Screening cost off**: $\gamma^x_{s,j} \to \infty$, so $s^x_j(z^x, S) = 1$ wherever the signal has value; errors remain if $\sigma_j < 1$. **Screening off**: also $\sigma_j = 1$, so by (40) a partner survives iff favorable; or $\mathcal{O} = \{\omega, \tau\}$.

Each switch is a limit of the same model, not a new model. With both stages off, a meeting becomes a match iff both accept (built below): the benchmark.

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- **Redraws off**: $\theta \to 0$. Every match then ends at the exogenous rate: $\delta(z^{\male}, z^{\female}) = \delta_0$.

Each switch is a limit of the same model, not a new model. With both stages off, a meeting becomes a match iff both accept (built below): the benchmark.

Values and Equilibrium

Value Functions (1/2): Matched ▷ Steps

$$\left[\beta + \delta_0 + \theta + \theta\,\Upsilon(\underline{y}^{-x})\right] W^x(z^{-x}; z^x \mid y) = \underbrace{u(z^{-x}; z^x \mid y)}_{\text{flow now}} + \underbrace{\delta_0\, U^x(\varnothing; z^x)}_{\text{exogenous end}} + \underbrace{\theta\,\Upsilon(\underline{y}^{-x})\, U^x(\varnothing; z^x)}_{\text{partner redraws, leaves}}$$
$$+ \underbrace{\theta \int \max\left\{W^x(z^{-x}; z^x \mid y'),\ U^x(\varnothing; z^x)\right\} d\Upsilon(y')}_{\text{own redraw: stay or leave}} \qquad (21)$$

$1/r^x$: how long the current draw keeps paying, discounted (until a δ_0 shock, x's own redraw, or the partner leaves). Redraws off: $W^x = [u + \delta_0 U^x]/(\beta + \delta_0)$.

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- $W^x(z^{-x}; z^x \mid y)$: the **continuation value of the match** to z^{-x} at draw y. $U^x(\varnothing; z^x)$: of being **unmatched** (next).

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Result (E2) **[proved]**

W^x is linear in y: $W^x(z^{-x}; z^x \mid y) = U^x(\varnothing; z^x) + \left(y - \underline{y}^x\right)/r^x$, $r^x \equiv \beta + \delta_0 + \theta + \theta\,\Upsilon(\underline{y}^{-x})$.

$1/r^x$: how long the current draw keeps paying, discounted (until a δ_0 shock, x's own redraw, or the partner leaves). Redraws off: $W^x = [u + \delta_0 U^x]/(\beta + \delta_0)$.

Value Functions (2/2): Unmatched

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$$\times \underbrace{\mathbb{P}(z^{\male}, z^{\female}, S)}_{\text{meeting}\to\text{match}} \underbrace{\left[W^x(z^{-x}; z^x \,|\, 0) - U^x(\varnothing; z^x|S)\right]}_{\text{gain from matching, at } y=0} \tag{22}$$

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- A new match starts at $y = 0$. **Search value**, a flow in the units of f, g and y:

$$V^x(z^x) \equiv \beta\, U^x(\varnothing; z^x) - b(z^x) \;\geq\; 0 \tag{24}$$

The Reservation Shock

Setting $W^x(z^{-x}; z^x \mid \underline{y}^x) = U^x(\varnothing; z^x)$ in (21):

$$\underbrace{\underline{y}^x}_{\substack{\text{the draw at}\\ \text{which } x \text{ leaves}}} + \underbrace{\frac{\theta}{r^x}\,\mathbb{E}\big[(y' - \underline{y}^x)^+\big]}_{\substack{\text{option to wait}\\ \text{for a better draw}}} = \underbrace{V^x(z^x) - v^x}_{\substack{\text{search's flow beyond}\\ \text{this partner's type}}} \tag{25}$$

$\delta = \delta(z^{\male}, z^{\female})$; partner's exit held fixed in the derivative. A partner likelier to leave shortens the time a good draw can pay, so waiting is worth less. Redraws off: $\underline{y}^x = V^x - v^x$.

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Result (E3) **[proved]**

Given the partner's exit probability, (25) has exactly one solution. It rises with $V^x(z^x)$ and falls with v^x, each more than one-for-one, $\partial \underline{y}^x / \partial V^x = r^x/(\beta + \delta) \geq 1$, and it **rises with the partner's exit probability** $\Upsilon(\underline{y}^{-x})$: leaving is a strategic complement.

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Result (E4) **[proved]**

With β, θ, Υ common to the sexes, the partner with the **smaller flow surplus** $v^x - V^x(z^x)$ has the higher reservation and **leaves more often**. Equal surpluses, equal reservations.

Several pairs, in every computed case, only when redraws are much faster than discounting: partners then coordinate on staying, or on leaving (backup). Shock scales by sex: compare $(v^x - V^x)/\sigma_y^x$.

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- **The comparison is cardinal**: his $f(z^{♀}; z^{♂}) - V^{♂}(z^{♂})$ against her $g(z^{♂}) - V^{♀}(z^{♀})$, in the units of y. With his own type he gets f_0; she gets g of her partner's type.

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- The pair $(\underline{y}^{♂}, \underline{y}^{♀})$ **always exists**; it is **unique** if the slopes of the two responses multiply to less than 1:

$$\prod_x \theta\, \Upsilon'(\underline{y}^{-x}) \left[V^x - v^x - \underline{y}^x\right] < \left[\beta + \delta(z^{♂}, z^{♀})\right]^2 \tag{26}$$

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The Surplus at Formation

- **Surplus at formation**, Δ^x: the gain from starting the match, at $y = 0$, rather than walking away. By the linearity of W^x,

$$\Delta^x(z^{-x}; z^x) \equiv W^x(z^{-x}; z^x \mid 0) - U^x(\varnothing; z^x) = \frac{-\underline{y}^x}{r^x}$$
$$= \frac{1}{r^x}\Big[\underbrace{v^x - V^x(z^x)}_{\text{payoff vs. search}} + \underbrace{\tfrac{\theta}{r^x}\mathbb{E}\big[(y' - \underline{y}^x)^+\big]}_{\text{option to redraw}}\Big] \qquad (27)$$

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- A partner **likelier to leave** lowers it: r^x rises and the option is worth less.

Redraws off: $\Delta^x = (v^x - V^x)/(\beta + \delta_0)$. The rate in the second bullet holds the partner's exit probability fixed.

Who Accepts Whom — The Decision After Contact & Screening

- **Accept iff the surplus at formation is positive**, i.e. iff the starting draw 0 clears the reservation. A^x is that **decision** (1 yes, 0 no); any value in $[0,1]$ is optimal at a tie:

$$A^{\male}(z^{\female}; z^{\male}) = \mathbf{1}\{\underline{y}^{\male} < 0\} = \mathbf{1}\Big\{f(z^{\female}; z^{\male}) + \tfrac{\theta}{r^{\male}}\mathbb{E}[(y')^+] > V^{\male}(z^{\male})\Big\} \tag{28}$$

$$A^{\female}(z^{\male}; z^{\female}) = \mathbf{1}\{\underline{y}^{\female} < 0\} = \mathbf{1}\Big\{g(z^{\male}) + \tfrac{\theta}{r^{\female}}\mathbb{E}[(y')^+] > V^{\female}(z^{\female})\Big\} \tag{29}$$

Redraws off: $A^x = \mathbf{1}\{v^x > V^x\}$. A partner likelier to leave raises r^x and is accepted less readily. For $\Upsilon = \text{Normal}(0, \sigma_y^2)$, $\mathbb{E}[(y')^+] = \sigma_y/\sqrt{2\pi}$.

Who Accepts Whom — The Decision After Contact & Screening

- **Accept iff the surplus at formation is positive**, i.e. iff the starting draw 0 clears the reservation. A^x is that **decision** (1 yes, 0 no); any value in $[0,1]$ is optimal at a tie:

$$A^{\male}(z^{\female}; z^{\male}) = \mathbf{1}\{\underline{y}^{\male} < 0\} = \mathbf{1}\Big\{f(z^{\female}; z^{\male}) + \frac{\theta}{r^{\male}}\mathbb{E}[(y')^+] > V^{\male}(z^{\male})\Big\} \tag{28}$$

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One rule starts and ends a match. The right to leave later **lowers the bar**: x accepts some partners worth less than search now, on the chance of a better draw.

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Four Decisions: Three in a Meeting, One in the Match

Each decision weighs a cost draw, a payoff or a draw against what continuing to search is worth:

- **Contact**: commit iff $c^x \leq \bar{c}^x(z^x, S)$, the expected surplus at formation over the partners x may meet in S, given what $\mathcal{O}$ shows:

$$\bar{c}^x(z^x, S) = \mathbb{E}\left[\Delta^x \mathbf{1}\{\text{match forms}\}\right] \tag{30}$$

Δ^x: the surplus at formation (27). The contact threshold does not condition on the partner's commitment: optimal when that commitment does not depend on hidden types, a simplification under pooling.

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- **Screen** on each hidden j: iff $\kappa_j^x \leq \Psi_j^x(z^x, S)$, what the signal adds:

$$\Psi_j^x(z^x, S) = \underbrace{\mathbb{E}\left[\Delta^x \mathbf{1}\{\text{accept on signal}\}\right]}_{\text{decide on the signal}} - \underbrace{\max\left\{\mathbb{E}[\Delta^x],\, 0\right\}}_{\text{decide blind}} \tag{31}$$

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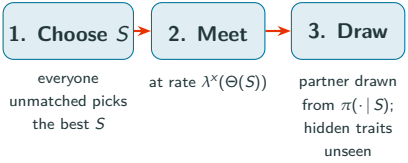
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- **Stay**: at each own redraw y', stay iff $y' \geq \underline{y}^x(z^{\male}, z^{\female})$.

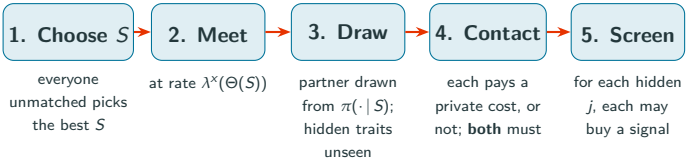
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The Path of a Meeting ▷ Known ▷ $\mathbb{P}$ Eq'm ▷ Def. 3.1′



Contact free and nothing hidden: the first two factors equal 1, and $\mathbb{P}(z^{\male}, z^{\female}, S) = A^{\male}(z^{\female}; z^{\male})\, A^{\female}(z^{\male}; z^{\female})$.

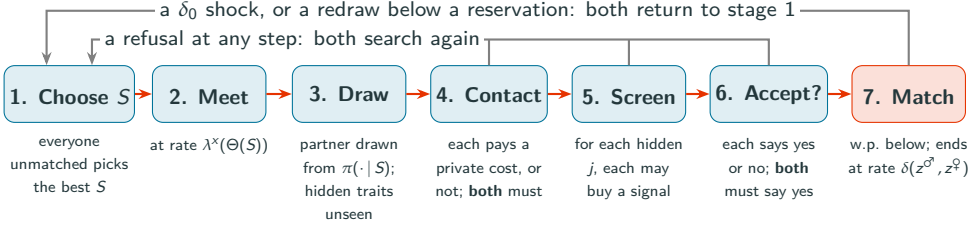
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The Path of a Meeting

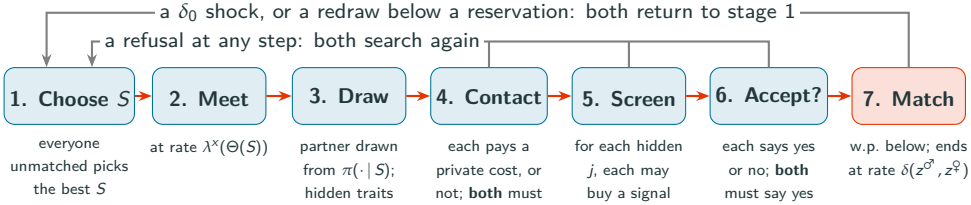
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The Path of a Meeting

▷ Known ▷ $\mathbb{P}$ Eq'm ▷ Def. 3.1′



- The probability a match between $z^{\male}$ and $z^{\female}$ in S:

$$\mathbb{P}(z^{\male}, z^{\female}, S) = \underbrace{P^{\male}(z^{\male}, S)\, P^{\female}(z^{\female}, S)}_{\text{both commit to contact}} \times \underbrace{\prod_{j \notin \mathcal{O}} P^{\male}_{s,j}(z^{\male}, z^{\female}, S)\, P^{\female}_{s,j}(z^{\female}, z^{\male}, S)}_{\text{both survive the screens}} \times \underbrace{A^{\male}(z^{\female}; z^{\male})\, A^{\female}(z^{\male}; z^{\female})}_{\text{both accept}} \tag{32}$$

Contact free and nothing hidden: the first two factors equal 1, and $\mathbb{P}(z^{\male}, z^{\female}, S) = A^{\male}(z^{\female}; z^{\male})\, A^{\female}(z^{\male}; z^{\female})$.

Benchmark — Perfect Information & Free Contact

The Benchmark Problem — Perfect Info., No Contact Costs



contact free, nothing hidden

- **One friction**: meeting takes time. Contact is free, nothing is screened, and every type is visible, $\mathcal{O} = \{\omega, \tau\}$. Matches end at δ_0 shocks and at redraws, as set out above.

The Benchmark Problem — Perfect Info., No Contact Costs



contact free, nothing hidden

- **One friction**: meeting takes time. Contact is free, nothing is screened, and every type is visible, $\mathcal{O} = \{\omega, \tau\}$. Matches end at δ_0 shocks and at redraws, as set out above.
- **Populations and type mixes**, in the order considered:
  1. **Baseline**: $N^{♂} = N^{♀}$, $\phi^{♂}(z) = \phi^{♀}(z) = \frac{1}{4}$ for every z, uniform (Prop. 3.1).
  2. **Unequal type mixes**: $N^{♂} = N^{♀}$, $\phi^{♂} \neq \phi^{♀}$ (Prop. 3.2).
  3. **More men**: $N^{♂} = \nu N^{♀}$, $\nu > 1$, same type fractions for both sexes (Prop. 3.3).

The Benchmark: Everyone Meets at the Same Rate ▷ Numerical ▷ 8 People

Result (Prop. 3.1′) [proved; normal Υ]

With everything visible, contact free, and equal numbers of each type on both sides, there is an equilibrium in which each type matches only its own type, and in it:

- every sub-market is balanced and every type forms matches at rate ξ; no one is excluded;
- each pair's hazard **falls with** g: every man gets f_0, so pairs differ only in what she gets;
- so time partnered, $\xi/(\xi + \delta(z, z))$, **rises with** g: the bottom's matches end soonest;
- women's welfare is more dispersed than b, since better women match better men.

Bullet 2: at a regular root of the exit decisions. Takeaway: when b ranks types as g does; first order in θ, and in every computed cell. Redraws off: Prop. 3.1. Refined if penalties are large enough. ▷ Moen

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The zero point is gone: $\mathrm{Var}(u^{\male}) > \mathrm{Var}\{b^{\male}\}$ even here, through how long matches last, not how often they form.

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Result (Prop. 3.2′) **[proved]**

Unequal type mixes **ration**: the surplus types match less often, but no type's match rate is zero. Redraws do not change how many are left over: in each sub-market the unmatched women exceed the unmatched men by the gap in type mixes, whatever the hazards.

Regressive case proved with redraws off (Prop. 3.2(iii)). With redraws, in the computed example the scarce men's matches also last longer, and the case stays regressive (backup).

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What Unequal Mixes Do ▷ Example ▷ Prop. 3.2′

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  - Every type $z^{♂}$ accepted but women meet him less often

Women out-educate men — high-ω men scarce, low-ω men in surplus — the regressive case.

Regressive case proved with redraws off (Prop. 3.2(iii)). With redraws, in the computed example the scarce men's matches also last longer, and the case stays regressive (backup).

Result (Prop. 3.3′)

Under sorting the excess men are **no longer spread evenly**: the sub-markets whose matches last longest — with redraws, the top — are the tightest, and their men are matched more of the time. Tight sub-markets make a cross sub-market pay: a high man accepts a lower-type woman to match sooner (he trades down in match quality), she gets a man above her type (she trades up), and the low men left behind face the tightest remaining sub-market.

First sentence proved given the hazards; that the top's hazards are lowest, computed (backup examples). Redraws off: every sub-market equally tight and every man matched the same share (Prop. 3.3(i)).

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- That is the pattern in the data: the poorest and least desirable men face the most dating difficulty (Hudson and den Boer 2004; Edlund et al. 2013).
- But with everything visible, matches for the bottom become less frequent; they do not disappear.

First sentence proved given the hazards; that the top's hazards are lowest, computed (backup examples). Redraws off: every sub-market equally tight and every man matched the same share (Prop. 3.3(i)).

What Search Frictions Alone Do

- **Equal numbers of each type**: every type forms matches at rate ξ, but women leave the men they value less sooner: the bottom is partnered least, and the market adds male inequality.
- **No type is excluded**: under sorting every type matches, for any N^x and ϕ^x of full support. Waits rotate within a type, so there are no celibates.
- **Unequal ϕ^x**: men scarce relative to the women of their type match more of the time; in the computed example their matches also last longer.
- **More men**: the longest-lasting sub-markets are the tightest; every woman gains.

Search frictions alone **ration**. They do not **exclude**.

Assumes every mismatch penalty is large enough that sorting passes the market-utility test (backup bound). Third and fourth bullets: computed examples.

Contact Costs with Perfect Information

Faster Matching, More Break-ups



contact costly, nothing hidden, valuations redrawn

In a sub-market pairing $z^{\male}$ with $z^{\female}_{\mp}$, each partner's reservation shock and the hazard are those of the model section (eqs. (25), (9)), with $v^{\male} = f(z^{\female}_{\mp}; z^{\male})$ and $v^{\female}_{\mp} = g(z^{\male})$.

Result (E5) **[proved; normal Υ]**

A faster match rate m raises both search values, so $\delta(z^{\male}, z^{\female}_{\mp})$ rises with m; at a given m, it falls with each valuation. Contact costs move δ only through m: cheaper contact for either sex raises every pair's match rate **and** its hazard.

$V^x = m\,\Delta^x$, with Δ^x the surplus at formation. At a regular root of the pair's exit decisions (always if $\theta < 3.7(\beta + \delta_0)$). Computed under sorting: every match rate and every hazard rose at each step of the cost of contact.

Result (Prop. 3.4′) **[proved; normal ϒ]**

With everything visible and costly contact, sorting remains an equilibrium, every sub-market is balanced, and every matched man gets f_0 from his partner's type. Match rates rise with g. With $\tilde{u} \equiv \overline{m}\,(f_0 + \mathbb{E}[y^{♂} \mid \text{matched}])$, the match part of a man's welfare,

$$\mathrm{Var}(u) = \mathrm{Var}\{b\} + \mathrm{Var}\{\tilde{u}\} + 2\,\mathrm{Cov}\{\tilde{u}, b\} \;\geq\; \mathrm{Var}\{b\}. \tag{33}$$

$\overline{m}$: the share of time partnered. The inequality holds when $\tilde{u}$ and b rank the types alike, and held in every computed cell; strict unless women's contact saturates for every type and redraws end no match. Redraws off: $\tilde{u} = f_0\,\overline{m}$, Prop. 3.4.

Result (Prop. 3.4′) **[proved; normal ϒ]**

With everything visible and costly contact, sorting remains an equilibrium, every sub-market is balanced, and every matched man gets f_0 from his partner's type. Match rates rise with g. With $\tilde{u} \equiv \overline{m}\,(f_0 + \mathbb{E}[y^{♂} \mid \text{matched}])$, the match part of a man's welfare,

$$\mathrm{Var}(u) = \mathrm{Var}\{b\} + \mathrm{Var}\{\tilde{u}\} + 2\,\mathrm{Cov}\{\tilde{u}, b\} \;\geq\; \mathrm{Var}\{b\}. \tag{33}$$

The **incidence wedge**: desirable men match more often, even with full transparency.

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Cheaper Contact Can Lower Partnership ▷ Example

Result (E6) **[criterion proved]**

With everything visible, cheaper contact raises a pair's match rate m **and** its hazard δ (E5). Its time partnered, $\overline{m} = m/(\delta + m)$, falls iff

$$\frac{d \ln \delta}{d \ln m} > 1 : \qquad \text{the hazard rises faster than the match rate.} \tag{34}$$

- Never without redraws. It needs break-ups driven by redraws, each of which rarely ends a match: a small rise in the reservation shocks then multiplies the hazard.

With everything visible, cheaper contact speeds every match **and** every break-up: partnership can fall with nothing hidden.

Computed under sorting: the share partnered falls over part of the range of contact costs in 224 of 384 cost paths at $\theta = 1$, $\sigma_y = 0.5$, all with $\delta_0 = 0.05$; never at $\theta = 0.1$.

Turnover and Hegemony with Perfect Information

Choose S → Meet → Draw → Contact → Screen → Accept contact costly; matches dissolve at rate $\delta(z^{\male}, z^{\female})$

Lemma 3.1′ **[proved]**

A male type forming matches at rate $m(z^{\male})$, whose matches end at hazard $\overline{\delta}(z^{\male})$, has unmatched mass

$$n^{\male}(z^{\male}) = \frac{\overline{\delta}(z^{\male})\, N^{\male}\, \phi^{\male}(z^{\male})}{\overline{\delta}(z^{\male}) + m(z^{\male})} = \left[1 - \overline{m}(z^{\male})\right] N^{\male}\, \phi^{\male}(z^{\male}). \tag{35}$$

- Searchers under-represent the types partnered most of the time (high $m/\overline{\delta}$), not simply the fastest to match; a type that never matches is over-represented by $(\overline{\delta} + m)/\overline{\delta}$.
- Once attributes are hidden, a woman draws from these men, not from the population.

$\overline{\delta}(z^{\male})$: his pairs' hazards, a harmonic mean weighted by formation rates. $\overline{m}(z^{\male})$: his share of time partnered.

Measuring Concentration: Two Indices

- **Flow index** (Def. 3.2): new matches per man of the type, over the average man's,

$$H_{\text{flow}}(z) = \frac{\overline{\delta}(z)\,\overline{m}(z)}{\sum_{z'} \phi^{\male}(z')\,\overline{\delta}(z')\,\overline{m}(z')}. \tag{36}$$

- **Time index**: the share of time partnered, over the average,

$$H_{\text{time}}(z) = \frac{\overline{m}(z)}{\sum_{z'} \phi^{\male}(z')\,\overline{m}(z')}. \tag{37}$$

- They coincide iff every type's matches end at the same rate: when $\theta = 0$, not otherwise. A type that changes partners often can lead on the first and trail on the second.
- **Sexual hegemon** (Def. 3.2): not a celibate, and $H_{\text{flow}}(z) > (1 - I)^{-1}$, I the celibates' share of men; H_{time} reported beside it.

By Lemma 3.1′ the flow of new matches to a type is $n^{\male}\, m = N^{\male}\, \phi^{\male}\, \overline{\delta}\, \overline{m}$. With everything visible, the women of each type have the indices of the men they match, on both measures.

Result (Prop. 3.5′) **[proved; third bullet partly computed]**

- H_{time} rises with $m/\overline{\delta}$ alone. H_{flow} rises with both m and $\overline{\delta}$: new matches per man are $(1/m + 1/\overline{\delta})^{-1}$.
- Time shares even out only if **both** $\delta_0 \to 0$ and $\theta \to 0$. Flow shares then tend to the types' relative hazards, not to equality.
- Everything visible: as contact becomes prohibitive, the top pairs lead on both indices; with contact free, the top leads on time and the bottom on new matches.

Limits at match rates bounded away from zero, as with everything visible. Between the two ends of the third bullet, computed: the top leads on time in every cell except some with fast redraws and intermediate contact costs (the bottom leads in 128 of 1,920 at $\theta = 1$); the flow leader moves from the top to the bottom as contact cheapens.

Contact Costs and Hegemony

Result (Props. 3.1′, 3.4′, 3.5′; E5): everything visible

- **Free contact**: every man forms matches at rate ξ, but the less a woman values him, the sooner his match ends: the top is partnered most, the bottom changes partners most.
- **Costly contact**: her contact threshold $\overline{c}^{\varphi}(z^{\varphi}, S)$, her expected surplus at formation, is larger with a more desirable man, so desirable men match faster.
- **Turnover** turns unequal rates and hazards into unequal shares, rejecting no one; time shares even out only as $\delta_0, \theta \to 0$.
- **Raising the cost of contact** (γ^x down) lowers every match rate and every hazard. Match rates are equal at both ends, unequal in between; time partnered stays unequal even with contact free.

Contact costs make match rates unequal; redraws make durations unequal. Either, with turnover, concentrates time partnered at the top.

Takeaway: in every computed cell but some with fast redraws and intermediate contact costs, where the top's faster matching brings faster break-ups.

Hidden Attributes and Screening

Hiding an Attribute Pools the Sub-market ▷ Decision



attributes hidden, screening on

- With j hidden ($j \notin \mathcal{O}$), men who differ on j search in **one sub-market**.
- A woman knows the sub-market's composition, not the specific type of the person she encounters.
- The screening stage switches on: before deciding, each side may buy a noisy signal on j, and a type favorable on j passes the other side's screen more often than one that is not.

Why Screening Creates Adverse Selection (♀ Perspective)

▷ Loop ▷ Several Eq.

1. **She draws from the unmatched**, not the population, and a type partnered most of the time spends little time unmatched (Lemma 3.1′).
2. **Without a screen** she treats the men she cannot tell apart alike.
3. **With a screen** the favorable survive more often, match faster, and **leave the pool first**.
4. **Redraws add a channel**: she leaves a less desirable man sooner, so he **returns to the pool sooner**.
5. The pool left behind is **tilted toward the less desirable** — worse than the population.

Computed, no one rejected: the higher man is under-represented in 95–97% of pooled sub-markets at $\theta = 0.1$; men's own exits push the other way.

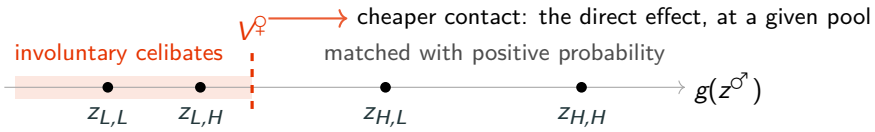
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5. The pool left behind is **tilted toward the less desirable** — worse than the population.

Adverse selection by exit: she faces a worse pool, not because bad types enter, but because good types are taken first, and kept.

Computed, no one rejected: the higher man is under-represented in 95–97% of pooled sub-markets at $\theta = 0.1$; men's own exits push the other way.

Who Matches: Celibates ▷ Cutoff ▷ On an App



Result (Prop. 3.6′) [proved; probation computed]

- A man with $g(z^{\male}) + \frac{\theta}{r^{\female}}\mathbb{E}[(y')^+] < V^{\female}$ is rejected by every woman who learns his type: he matches only by **screening error**.
- Celibates fill from the bottom of g; none with everything visible. **No woman is a celibate**: men of her own type accept her.
- **Redraws lower the bar**: marginal men are accepted **on probation** and left sooner.

Def. 3.3′: rejected at formation by every woman of his sub-market who learns his type. Order up to a narrow band. Computed (τ-directed): men held at the cutoff in 340 steady states at $\theta = 0$, 32 at $\theta = 0.1$; on probation, partnered 0.89 as much as the others.

Contact Costs and Women's Cutoffs ▷ Figure

Result (Prop. 3.6(iii)′, Cor. A.1): attributes hidden

- **Direct effect, at a given pool.** Cheaper contact for women ($\gamma^{\text{♀}}$ up) raises the value of continuing to search, so women's cutoffs rise. Types below the new cutoff become celibates; types above match at least as often. Costlier contact does the reverse: cutoffs fall and marginal men are readmitted.
- **Uncertainty drives cutoffs down.** Cheaper contact speeds the matching of every man she cannot tell apart. Those who pass her screen leave faster, the pool left behind is worse, and her search value falls. The total effect is not signed in general; the pool effect vanishes as matches end faster.
- **Men's response can reverse it.** Men who match faster can stop accepting mismatched women, and with redraws they also leave sooner, so each match is worth less to her.

Cheaper contact raises women's standards at a given pool; the pool's response and men's response push back.

Direct effect: men accept the sub-market's women alike, women learn types on contact. Proved with redraws off; computed with them (backup).

Turnover and Exclusion ▷ Computed

Result (§3.6 in paper; E7): attributes hidden

Two male types, $g_{lo} < g_{hi}$, share a sub-market, and the pool responds. With redraws off, the lower type is excluded outright iff

$$\frac{g_{hi}}{g_{lo}} > R^*(\delta_0) = 2 + \frac{2(\beta^{\varphi}_{\mathfrak{F}} + \delta_0)}{\lambda} + \frac{\beta^{\varphi}_{\mathfrak{F}}}{\delta_0}, \tag{38}$$

with λ the meeting rate; otherwise he is held at the cutoff. With redraws the threshold stays **finite as** $\delta_0 \to 0$: the accepted pair's break-ups keep returning men to the pool.

- **Lower continuation values readmit the bottom**: a given pool supports a lower cutoff.
- **The pool excludes it**: accepted men return to it sooner, so the rejected are less of it.
- The pool dominates when matches are durable, continuation values when they are short.

Types learned at the meeting; men accept every woman; contact free; the two types equally common. Finiteness proved. Computed: still U-shaped in δ_0, below R^* where δ_0 is small, above it where large.

Conclusion

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Conclusion

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- **Cheap contact speeds matching and break-ups alike**: with types visible it can lower partnership; with types hidden, at a given pool, it raises women's standards.
- **Break-ups soften exclusion into probation**: the marginal man is accepted and left early, partnered less rather than never.
- **Open:** which index defines a hegemon — new matches or time partnered; beliefs, the planner and remedies, not yet reassessed with break-ups.

Thanks!

Appendix Slides

Four Regimes: What Is Visible Before Meeting

Regime	Visible $\mathcal{O}$	Sub-markets	Random within
Full directed	$\{\omega, \tau\}$	up to 16	nothing
ω-directed (credentials verified)	$\{\omega\}$	up to 4	τ
τ-directed (photo first)	$\{\tau\}$	up to 4	ω
Random	$\varnothing$	1	ω and τ

- **Random search** is the same model with nothing visible: one sub-market, and the only choice is whether to search.
- Hidden attributes are **pooled**: agents who differ on them search in one sub-market. The other side knows its *composition*, but sees only the visible part of a partner's type.

Screening: Who Survives a Screen

- **Favorable** on j: for her, a man high on j; for him, a woman matching him on j. The **prior**:

$$q_j^x(z^x, S) = \sum_{z^{-x}} \pi(z^{-x}|S)\,\mathbf{1}\{z^{-x} \text{ favorable on } j \text{ for } z^x\} \tag{39}$$

- **Unscreened**, x accepts every partner iff $q_j^x(z^x, S) \geq \widehat{q}_j^x(z^x, S)$, the prior at which accepting blind breaks even (derived later).
- **Survival** on j: x accepts blind without screening, or screens and the signal reads favorable:

$$\begin{aligned} P_{s,j}^x(z^x, z^{-x}, S) = {} & \left[1 - s_j^x(z^x, S)\right]\mathbf{1}\left\{q_j^x(z^x, S) \geq \widehat{q}_j^x(z^x, S)\right\} \\ & + s_j^x(z^x, S) \times \begin{cases} \sigma_j & \text{if } z^{-x} \text{ is favorable on } j \\ 1 - \sigma_j & \text{if not} \end{cases} \end{aligned} \tag{40}$$

The favorable survive more often, by $s_j^x(z^x, S)\,(2\sigma_j - 1)$: a hidden difference in type becomes a difference in match rates.

What Is Known at Each Stage

◁ Path

- **After stage 3**, the pair is known: the probability is (32). With contact free and nothing hidden, it is 1 or 0 away from a tie.
- **After stage 2**, a meeting has happened but the partner is unknown. Averaging over the draw in stage 3, the probability that the meeting ends in a match is

$$\overline{\mathbb{P}}^{\male}(z^{\male}, S) = \sum_{z^{\female}} \pi(z^{\female} \mid S)\, \mathbb{P}(z^{\male}, z^{\female}, S) \tag{41}$$

$$\overline{\mathbb{P}}^{\female}(z^{\female}, S) = \sum_{z^{\male}} \pi(z^{\male} \mid S)\, \mathbb{P}(z^{\male}, z^{\female}, S) \tag{42}$$

- **After stage 1**, no one has been met, so there is a rate, not a probability: that type matches in S at rate $\lambda^{x}(\Theta(S))\, \overline{\mathbb{P}}^{x}(z^{x}, S)$.
- **Before stage 1**: averaging that rate over $S^{*}(S \mid z^{x})$ gives the type's match rate (below).

$\mathbb{P}$ Is an Equilibrium Object (1/2) ◁ Path

1. $\mathbb{P}$ **depends on search values**: each A^x compares a payoff, f or g, plus the option value of a redraw, with $V^x(z^x)$.

2. **Search values depend on the market**:

$$V^{♂}(z^{♂}) = \max_{S} \lambda^{♂}(\Theta(S)) \sum_{z^{♀}} \pi(z^{♀}|S)\, \mathbb{P}(z^{♂}, z^{♀}, S)\, \Delta^{♂}(z^{♀}; z^{♂}) \tag{43}$$

$$V^{♀}(z^{♀}) = \max_{S} \lambda^{♀}(\Theta(S)) \sum_{z^{♂}} \pi(z^{♂}|S)\, \mathbb{P}(z^{♂}, z^{♀}, S)\, \Delta^{♀}(z^{♂}; z^{♀}) \tag{44}$$

3. **The market depends on who enters**: the entry shares $S^*(S|z^x)$ give

$$n^x(z^x|S) = S^*(S|z^x)\, n^x(z^x), \tag{45}$$

and these masses set $\Theta(S)$ and $\pi(\cdot \mid S)$.

$\mathbb{P}$ Is an Equilibrium Object (2/2) ◁ Path

4. **Who is unmatched depends on** $\mathbb{P}$: each type's match rate and unmatched mass are

$$m(z^{\male}) = \sum_S S^*(S|z^{\male})\,\lambda^{\male}(\Theta(S))\,\overline{\mathbb{P}}^{\male}(z^{\male}, S), \qquad n^{\male}(z^{\male}) = \frac{\overline{\delta}(z^{\male})N^{\male}\phi^{\male}(z^{\male})}{\overline{\delta}(z^{\male}) + m(z^{\male})} \tag{46}$$

$$m(z^{\female}) = \sum_S S^*(S|z^{\female})\,\lambda^{\female}(\Theta(S))\,\overline{\mathbb{P}}^{\female}(z^{\female}, S), \qquad n^{\female}(z^{\female}) = \frac{\overline{\delta}(z^{\female})N^{\female}\phi^{\female}(z^{\female})}{\overline{\delta}(z^{\female}) + m(z^{\female})} \tag{47}$$

5. **The hazards depend on search values**: each reservation shock $\underline{y}^x$ compares a draw with $V^x(z^x) - v^x$, and $\overline{\delta}$ averages the pair hazards (Lemma 3.1′).

$\mathbb{P}(z^{\male}, z^{\female}, S)$, $V^x(z^x)$, $\underline{y}^x$, $\delta(z^{\male}, z^{\female})$, $S^*(S|z^x)$, $n^x(z^x|S)$, $\Theta(S)$ and $\pi(\cdot \mid S)$ are **solved jointly**. None is given.

Equilibrium (Def. 3.1′) ◁ Path

Competitive search equilibrium (Def. 3.1′)

1. Values solve the equations for $W^x(z^{-x}; z^x \mid y)$ and $U^x(\varnothing; z^x)$, each agent searching in the best sub-market.
2. Sub-market choice, contact, screening, acceptance and staying follow the four decision rules; the reservation shocks $\underline{y}^x$ solve (25) for each pair.
3. Active sub-markets' $n^x(S)$, $\Theta(S)$, $\pi(\cdot \mid S)$ are what choices and match flows generate.
4. Steady state: new matches of each type pair $= \delta(z^{\male}, z^{\female}) \times$ the mass of such matches.

Redraws off: Def. 3.1, with every hazard equal to δ_0.

The Matched Value, Step by Step

◁ Values

1. The integral in (21) does not depend on the current y; nor does the partner's exit rate, since the partner leaves only at a redraw. So W^x is linear in y, with slope $1/r^x$.
2. With $\Delta^x(y) \equiv W^x(z^{-x}; z^x|y) - U^x(\varnothing; z^x)$, subtract $r^x U^x$ from (21) and use $V^x = \beta U^x - b$:
$$r^x \Delta^x(y) = v^x - V^x(z^x) + y + \theta\, \mathbb{E}\left[\Delta^x(y')^+\right] \tag{48}$$
3. Linearity and $\Delta^x(\underline{y}^x) = 0$ give $\Delta^x(y) = (y - \underline{y}^x)/r^x$; at $y = \underline{y}^x$ this is (25).
4. The left side of (25) has slope $(\beta + \delta)/r^x > 0$ in $\underline{y}^x$: exactly one root.
5. A match ends at rate δ_0, $\theta\, \Upsilon(\underline{y}^{\male})$ through his redraws and $\theta\, \Upsilon(\underline{y}^{\female})$ through hers, none depending on its age: (9).

Nothing here depends on the form of f or g. Checked by a joint value iteration on both partners' draws that imposes none of this (`verify_endog_delta.py`, test 3).

Exit Can Coordinate: Two Stable Pairs

◁ Who Leaves

- Each partner's reservation rises with the other's exit probability: expectations about leaving can fulfil themselves.
- **When:** with Normal shocks and partners with equal surplus, several pairs exist for some $V^{x} - v^{x}$ iff $\theta/(\beta + \delta_0) > 17$ (about); σ_y only rescales $V^{x} - v^{x}$.
- **Example:** $\sigma_y = 1$, $\beta = 0.01$, $\delta_0 = 0$, $\theta = 10$, $V^{x} - v^{x} = -0.039$ for both:

	$\underline{y}^{\male} = \underline{y}^{\female}$	exit prob. per redraw	$\delta(z^{\male}, z^{\female})$	stable?
Both stay	-39.45	0.000	0.000	yes
Middle	-1.98	0.024	0.480	no
Both leave readily	-0.79	0.215	4.301	yes

One pair's exit decisions, whatever the preferences behind v^{x}. `agent_A_checks.py`, section E (endogenous-δ deck's records). A joint value iteration ends at the first pair from "both stay", at the third from "both leave".

Steady State with Pair-Specific Hazards (Lemma 3.1′) ◁ Hazard

- An unmatched $z^{\male}$ forms matches with $z^{\female}$ at rate $m(z^{\male}, z^{\female})$, in all $m(z^{\male}) = \sum_{z^{\female}} m(z^{\male}, z^{\female})$. His matches end at the harmonic-mean hazard

$$\frac{1}{\overline{\delta}(z^{\male})} = \sum_{z^{\female}} \frac{m(z^{\male}, z^{\female})}{m(z^{\male})} \frac{1}{\delta(z^{\male}, z^{\female})} \tag{49}$$

- Balancing each pair's flows, $n^{\male}(z^{\male})\, m(z^{\male}, z^{\female}) = \delta(z^{\male}, z^{\female}) \times$ its matched mass, gives

$$n^{\male}(z^{\male}) = \frac{\overline{\delta}(z^{\male})\, N^{\male}\, \phi^{\male}(z^{\male})}{\overline{\delta}(z^{\male}) + m(z^{\male})}, \qquad \text{time partnered } \overline{m}(z^{\male}) = \frac{m(z^{\male})}{\overline{\delta}(z^{\male}) + m(z^{\male})} \tag{50}$$

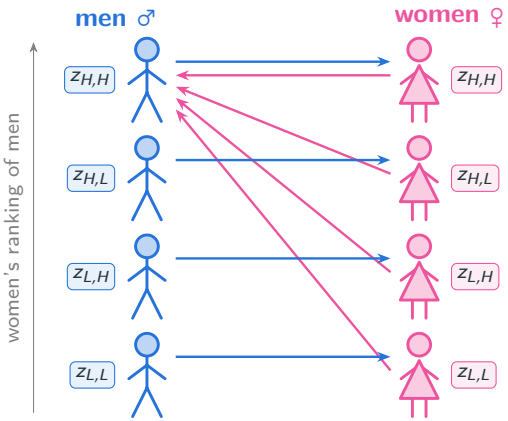
- Lemma 3.1 with δ replaced by $\overline{\delta}(z^{\male})$: types that match fast, **or whose matches last**, are scarce in the pool.

Pool share $\pi(z^{\male}) = n^{\male}(z^{\male}) / \sum_{z'} n^{\male}(z')$; within S, weight by $S^*(S|z^{\male})$. Women: the sexes exchanged. New matches to the type: $n^{\male}(z^{\male})\, m(z^{\male}) = \overline{\delta}(z^{\male})\, \overline{m}(z^{\male})\, N^{\male}\, \phi^{\male}(z^{\male})$.

A First Example: Eight People, Nothing Hidden

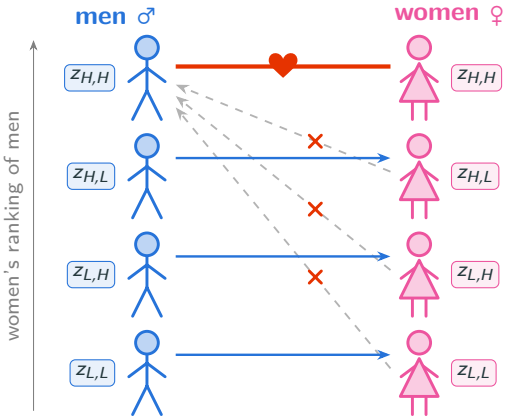
◁ Prop. 3.1

- One man and one woman of each type; everyone sees every type.
- **Women** (one ranking): all four want the $z_{H,H}$ man.
- **Men** (own kind): each wants the woman of his own type.



The Top Pair Forms ◁ Prop. 3.1

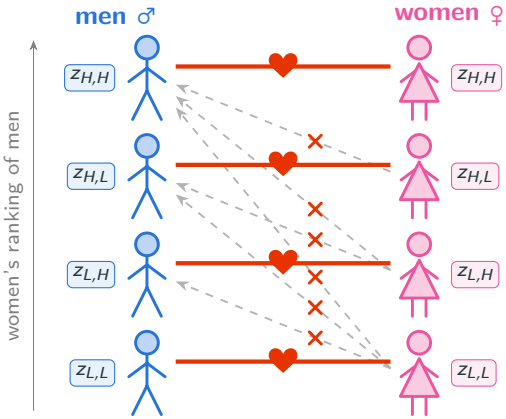
- The $z_{H,H}$ man is every woman's first choice; his is the $z_{H,H}$ woman, and he is hers. **They match.**
- The other three wanted him too, but they are not his kind, and he is taken.



The Same Logic, One Rung Down

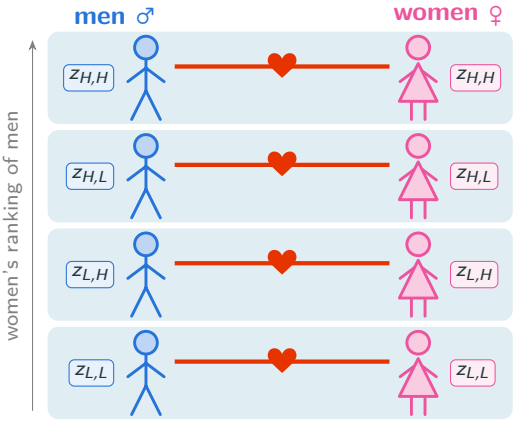
◁ Prop. 3.1

- The $z_{H,L}$ man is now the **best man left**; he wants the $z_{H,L}$ woman. **They match.**
- Then the $z_{L,H}$ pair, then the $z_{L,L}$ pair: **everyone ends up with their own type**.



Everyone Sees It Coming ◁ Prop. 3.1

- Every type is visible, so **everyone foresees this**.
- **Men** search for their own type: she is his first choice, and she says yes.
- **Women** do not search for better men: those men seek their own kind, so she would meet no one willing.
- ⇒ **Same-type sorting**: four active sub-markets, one per type.



A Numerical Example (Benchmark)

- $\omega \in \{1, 2\}$, $\tau \in \{0.5, 1\}$; a share $\frac{1}{4}$ each
- $g = b$; $f_0 = 1$, $d_\omega = d_\tau = 0.5$
- $\xi = 1$, $\beta = 0.05$, $\delta_0 = 0.05$
- $\theta = 0.5$, $\sigma_y = 0.5$

sub-market	$V^{\male}(z^{\male})$	$V^{\female}(z^{\female})$	$\delta(z, z)$	$\overline{m}$
$(z_{L,L}, z_{L,L})$	0.915	1.29	0.135	0.881
$(z_{L,H}, z_{L,H})$	0.929	1.72	0.121	0.892
$(z_{H,L}, z_{H,L})$	0.933	2.15	0.117	0.896
$(z_{H,H}, z_{H,H})$	0.934	2.57	0.116	0.896

- Everyone accepts at formation. His exits run at about 0.065 on every rung; hers fall from 0.021 at the bottom to 0.000 at the top: the ranking enters through her.
- Time partnered rises from 0.881 to 0.896, so $\mathrm{Var}(u) = 0.321 > 0.3125 = \mathrm{Var}\{b\}$. H_{time} runs from 0.988 (bottom) to 1.006 (top); H_{flow} from 1.096 to 0.954.

Redraws off ($\delta = 0.1$): $V^{\male} = 0.87$ for all, every man matched 91% of the time, $\mathrm{Var}(u) = \mathrm{Var}\{b\}$. Sorting survives the refinement in both. `dp_example.py`.

Is Sorting Robust? Empty Sub-markets

◁ Prop. 3.1

- Sorting leaves every cross sub-market (say, high men with low women) **empty**.
- Under Def. 3.1, a deviator to an empty sub-market **meets nobody** ($M = 0$), so no one deviates.
- That argument is too easy: it also sustains **anti-assortative** matching.
- **The market-utility test** (Moen 1997): could a man of type z and a woman of type $z' \neq z$ *both* do better in a sub-market of their own, at *some* tightness?

The Market-Utility Test, Step by Step

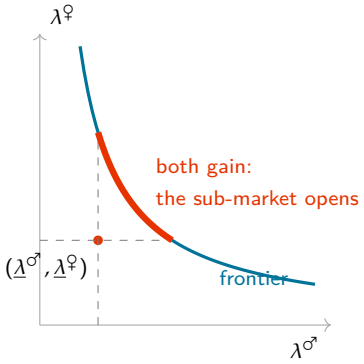
◁ Prop. 3.1

1. In a new sub-market each meets only the other's type, and gains Δ^x from each meeting: the pair's surplus at formation (27), at the current search values.
2. That beats the current search value $V^x(z^x)$ iff $\Delta^x > 0$ and the meeting rate λ exceeds the **break-even rate**

$$\underline{\lambda}^x = \frac{V^x(z^x)}{\Delta^x} \tag{51}$$

(redraws off: $(\beta + \delta_0)V^x/(\text{payoff} - V^x)$, payoff f for him, g for her).

3. Tightness trades the two rates along the **frontier** $(\lambda^{\male})^{1-\psi}(\lambda^{\female})^{\psi} = \xi$.
4. It **opens** iff both surpluses are positive and $(\underline{\lambda}^{\male})^{1-\psi}(\underline{\lambda}^{\female})^{\psi} < \xi$.



Drawn for $\psi = \frac{1}{2}$, $\xi = 1$.

When Is Sorting Safe? $\min_j d_j \geq T^*_\theta$

◁ Prop. 3.1′

1. A cross sub-market offers him $f_0 - d$, with d the mismatch penalty. By (28) he accepts it at formation only if $f_0 - d + \frac{\theta}{r^{\sigma}}\mathbb{E}[(y')^+] > V^{\sigma}(z^{\sigma})$, and $r^{\sigma} \geq \beta + \delta_0 + \theta$.
2. Under sorting his search value is at least what he would get by never leaving while every redraw of hers ends the match: $V^{\sigma}(z^{\sigma}) \geq \xi f_0/(\beta + \delta_0 + \theta + \xi)$.
3. So no cross sub-market opens if every penalty is at least the **bound**

$$T^*_\theta \equiv \frac{f_0\,(\beta + \delta_0 + \theta)}{\beta + \delta_0 + \theta + \xi} + \frac{\theta\,\mathbb{E}[(y')^+]}{\beta + \delta_0 + \theta'} \tag{52}$$

and the smallest penalty is $\min\{d_\omega, d_\tau\}$ (a double mismatch costs $d_\omega + d_\tau$).

T^*_θ: what a man would give up to match now rather than wait for his own type, plus what the right to leave adds to any new match. **Redraws raise it.**

Sufficient, not necessary: a cross sub-market also needs the woman to gain. Redraws off: $T^* = f_0(\beta + \delta_0)/(\beta + \delta_0 + \xi)$. Computed: sorting survived in every free-contact cell meeting the bound (504).

Below the Bound, Sorting Can Fail ◁ Prop. 3.1′

- Two types; $b = (2, 1)$, $g = (1.1, 1)$, $f_0 = 1$, $\xi = 1$, $\psi = \frac{1}{2}$, $\beta = 0.05$, $\delta_0 = 0.05$, $\theta = 0.5$, $\sigma_y = 0.5$: $T^*_\theta = 0.54$.
- $d = 0.5$, just below T^*_θ: **sorting survives**, and is the only refined steady state found.
- $d = 0.05$: high men and low women **open a sub-market of their own** — he gains speed, she trades up. Without redraws, at the same δ_0, sorting survives at $d = 0.05$: redraws raise the bar sorting must clear.
- High men are then partnered more of the time (0.865 vs. 0.848): $\mathrm{Var}(u) = 0.264$, against 0.255 under sorting and $\mathrm{Var}\{b\} = 0.25$.

`dp_twotype.py`; at both penalties an anti-assortative steady state also exists and fails the refinement. Four-type grid (`verify_endog_delta_basepref.py`): sorting survives in 184 of 192 free-contact cells at each θ.

How Numbers and Type Mixes Enter

◁ Prop. 3.1

- **Steady state**: each type's population is its unmatched plus its matched. N^x and ϕ^x enter only here, setting who is unmatched and so $\Theta(S)$:

$$N^x \phi^x(z^x) = n^x(z^x) + \sum_{S \ni z^x} \frac{M(S)\,\mathbb{P}(z^{\male}, z^{\female}, S)}{\delta(z^{\male}, z^{\female})} \tag{53}$$

- Under **perfect sorting** each type matches only its own. With $e(z)$ the mass of matched (z, z) pairs, each sub-market solves

$$\delta(z,z)\, e(z) = \xi \left[N^{\male} \phi^{\male}(z) - e(z)\right]^{1-\psi} \left[N^{\female} \phi^{\female}(z) - e(z)\right]^{\psi} \tag{54}$$

Case	Tightness	Who is unmatched most	Inequality among men
Equal N^x, ϕ^x	$\Theta(S) = 1$	the bottom: matches end sooner	through hazards only
Unequal ϕ^x	differs across S	over-represented types	can change
$N^{\male} > N^{\female}$	$\Theta(S) > 1$, highest at the top	the bottom men	through hazards and tight

A lower $\delta(z,z)$ raises $e(z)$ and, with more men, $\Theta(z,z)$. Redraws off: every $\delta(z,z) = \delta_0$.

Same Market, Different Mix

◁ Prop. 3.2

	high	low
men	0.5	0.5
women	**0.6**	**0.4**

- One dimension, everything visible, $N^{\male} = N^{\female} = 1$.
- $g = (1.1, 1)$, $b = (2, 1)$, $f_0 = 1$, $\xi = 1$, $\psi = \frac{1}{2}$, $\beta = 0.05$; $\delta_0 = 0.05$, $\theta = 0.5$, $\sigma_y = 0.5$; a small penalty $d = 0.05$.

- A surplus of 0.1 high women and, by arithmetic, 0.1 low men.
- Sub-markets are now unbalanced: tightness $\Theta \neq 1$ is the price, paid in waiting time.

Sorting: Rationing Without Exclusion ◁ Prop. 3.2

sub-market	Θ	$\delta(z,z)$	unmatched ♀ − ♂	matched ♂	matched ♀
(H,H)	0.32	0.184	$+0.10$	0.91	0.76
(L,L)	3.75	0.194	-0.10	0.73	0.91

- Sorting is still an equilibrium: high women wait ($\Theta < 1$), low men wait ($\Theta > 1$).
- **No one is excluded**: every type matches; 0.10 on each side is unmatched at any instant, *rotating* within the surplus types — the same gap as without redraws.
- The rationed men are the **worst-off**, and their matches also end sooner: $\mathrm{Var}(u) = 0.416 > 0.250 = \mathrm{Var}\{b\}$. Regressive — but at $d = 0.05$ it fails the refinement.

Redraws off ($\delta = 0.1$): $\Theta = 0.16$ and 8.57, matched men 0.96 and 0.77, $\mathrm{Var}(u) = 0.353$. `dp_twotype.py`.

Or the Surplus Is Absorbed: A Cross Sub-market

◁ Prop. 3.2

- Surplus high women search for surplus low men. It needs **two acceptances** at formation:

$$\underbrace{f_0 - d + \frac{\theta}{r^{\male}}\mathbb{E}[(y')^+] \geq V_L^{\male}}_{\text{he accepts the mismatch}} \qquad \underbrace{g(L) + \frac{\theta}{r^{\female}}\mathbb{E}[(y')^+] \geq V_H^{\female}}_{\text{she trades down}} \tag{55}$$

- At $d = 0.05$ it exists and **survives the refinement**: low men's matched share rises from 0.73 to 0.82. Still regressive: $\mathrm{Var}(u) = 0.312$.
- At $d = 0.5$ the men refuse: only sorting remains, and it survives.

V_L^x, V_H^x: the search value $V^x(z^x)$ of a low and a high type. Redraws off: matched share $0.77 \to 0.86$. `dp_twotype.py`.

Proposition 3.2′: Unequal Type Distributions

Maintain Proposition 3.1′, but with $N^{\male} = N^{\female}$ and $\phi^{\male} \neq \phi^{\female}$ of full support. A type is *over-represented* on a side if its mass there exceeds its mass on the other.

(i) Sorting remains an equilibrium and no type is excluded. Beyond the frictional unmatched, a mass $N^{\female} \sum_z [\phi^{\female}(z) - \phi^{\male}(z)]^+$ of women and an equal mass of men are unmatched at every instant, whatever the hazards, borne as lower matched shares by the over-represented types.

(ii) A cross sub-market in which over-represented men $z^{\male}$ absorb over-represented women $z^{\female}$ needs both to accept at formation, (28) and (29).

(iii) Redraws off: with one over-represented type per side, the burden is regressive, raising male inequality above that of $b(z)$, if the male type in deficit lies above and the one in excess below mean b (one strictly). It is progressive in the reverse case; otherwise it depends on the configuration.

(iv) Sorting survives the market-utility refinement if no mismatch pays any man more than search, counting the option value: $f_0 - \min_j d_j + \theta\, \mathbb{E}[(y')^+]/(\beta + \delta_0 + \theta) \leq V^{\male}(z^{\male})$ for every man. With smaller penalties the refinement can select absorption or trading up.

(i), (ii), (iv) proved with redraws; (iii) with redraws off, and the computed example stays regressive with them.

Same Types, More Men

◁ Prop. 3.3′

The same market (with redraws), $d = 0.05$, $g = (1.1, 1)$, half H and half L on both sides, but $N^{\male} = \nu N^{\female}$, $\nu > 1$.

Sorting steady state, H / L	$\nu = 1$	$\nu = 1.1$	$\nu = 1.2$
Θ	1 / 1	1.93 / 1.87	3.19 / 3.06
$\delta(z, z)$	0.162 / 0.172	0.168 / 0.177	0.180 / 0.188
matched share, men	0.860 / 0.853	0.811 / 0.805	0.757 / 0.752
matched share, women	0.860 / 0.853	0.892 / 0.885	0.909 / 0.903

- The high pairs last longer, so their sub-market is **tighter** and its men are matched more of the time: rationing without exclusion, but no longer neutral across types ($\mathrm{Var}(u) > \mathrm{Var}\{b\}$, by little here). Every woman gains.

Redraws off ($\delta = 0.1$): $\Theta = 2.76, 6.17$ in every sub-market, every man matched 0.86, 0.80 of the time (Prop. 3.3(i)). `dp_twotype.py`.

Tight Sub-markets Make Men Less Picky

◁ Prop. 3.3′

- A high man in a tight sub-market values **speed**; a low woman gains by **trading up**. Both can gain from a sub-market of their own.
- It opens when her proportional gain beats his proportional loss, in surpluses at formation, by enough to cover the gap in tightness:

$$\left(\frac{\Delta^{\female}(z_H; z_L)}{\Delta^{\female}(z_L; z_L)}\right)^{\psi}\left(\frac{\Delta^{\male}(z_L; z_H)}{\Delta^{\male}(z_H; z_H)}\right)^{1-\psi} > \left(\frac{\Theta(z_L, z_L)}{\Theta(z_H, z_H)}\right)^{\psi(1-\psi)}. \tag{56}$$

- With redraws the high sub-market is the tighter, so the right side is **below 1**: trading up opens more easily than with equally tight sub-markets.
- More men lower $V^{\male}(z^{\male})$ (men accept more) and raise $V^{\female}(z^{\female})$ (trading up pays more): the larger the surplus of men, the more easily the sub-market opens (redraws off, Prop. 3.3(ii)–(iii)).

Surpluses at the current search values. Redraws off: equal tightness, right side 1, and the surplus ratios are the payoff gaps $(g(H) - V_L^{\female})/(g(L) - V_L^{\female})$ and $(f_0 - d - V_H^{\male})/(f_0 - V_H^{\male})$.

Trading Up Pushes the Burden Down

◁ Prop. 3.3′

	$\nu = 1.1$		$\nu = 1.2$	
matched share	sorting	trading up	sorting	trading up
high men	0.81	0.83	0.76	0.80
low men	0.81	**0.78**	0.75	**0.71**
high women	0.89	0.88	0.91	0.90
low women	0.89	0.89	0.90	0.91
$\mathrm{Var}(u)$, with $\mathrm{Var}\{b\} = 0.25$	0.253	0.287	0.252	0.312

- High men move into the sub-market with low women; low men are left in the tightest sub-market: **regressive**, more so the larger the surplus.
- At $d = 0.05$ **only trading up survives** the refinement; at $d = 0.5$ sorting does.

$\delta_0 = 0.05$, $\theta = 0.5$, $\sigma_y = 0.5$. Redraws off ($\delta = 0.1$): low men 0.82, 0.71; $\mathrm{Var}(u) = 0.29$, 0.34.

Proposition 3.3′

Maintain Proposition 3.1′, but with $N^{\male} = \nu N^{\female}$, $\nu > 1$, and the same type fractions for both sexes.

(i) Sorting is an equilibrium, and no type is excluded. Every sub-market in use has $\Theta(S) > 1$; given the hazards, $\Theta(z, z)$ and the men's matched share are higher the lower is $\delta(z, z)$. Redraws off: every $\Theta(S)$ equal and increasing in ν, every man the same matched share, and male welfare dispersed exactly as much as $b(z)$.

(ii) Sorting survives the market-utility refinement iff no cross sub-market can be opened. The one pairing men z_i with women $z_j \neq z_i$ can be opened iff both surpluses at formation are positive and

$$\left(\frac{\Delta^{\female}(z_i; z_j)}{\Delta^{\female}(z_j; z_j)}\right)^{\psi} \left(\frac{\Delta^{\male}(z_j; z_i)}{\Delta^{\male}(z_i; z_i)}\right)^{1-\psi} > \left(\frac{\Theta(z_j, z_j)}{\Theta(z_i, z_i)}\right)^{\psi(1-\psi)} \tag{57}$$

which, with redraws off, requires $g(z_i) > g(z_j)$: she trades up, and he trades match quality for speed.

(iii) Redraws off: the left side of (57) rises in ν, the more surplus men, the more easily sorting is blocked.

Costly Contact: The Result

◁ Prop. 3.4′

Proposition 3.4′

Maintain Proposition 3.1′ (everything visible, equal numbers, uniform type mixes), and switch on the contact stage. Same-type sorting is an equilibrium. It survives the market-utility refinement whenever no mismatch pays any man more than his search value, counting the option value:
$f_0 - \min\{d_\omega, d_\tau\} + \theta\,\mathbb{E}[(y')^+]/(\beta+\delta_0+\theta) \le V^{\male}(z^{\male})$ for every male type. In it:

(i) Every matched man gets f_0 from his partner's type; his search value is $V^{\male}(z^{\male}) = m(z^{\male})\,\Delta^{\male}(z^{\male}; z^{\male})$.

(ii) $\Theta(S) = 1$ in every active sub-market, but $V^{\female}(z^{\female})$ rises strictly in g, and $m(z^{\male})$ is non-decreasing in g, strictly between any two types unless contact is saturated for both.

(iii) Each pair's hazard rises with its match rate and, at a given match rate, falls with g (E5).

(iv) With $\overline{m}(z^{\male})$ the share of time a man is matched and $\tilde{u} \equiv \overline{m}\,(f_0 + \mathbb{E}[y^{\male} \mid \text{matched}])$,

$$\mathrm{Var}(u) = \mathrm{Var}\{b\} + \mathrm{Var}\{\tilde{u}\} + 2\,\mathrm{Cov}\{\tilde{u}, b\} \;\ge\; \mathrm{Var}\{b\} \tag{58}$$

when $\tilde{u}$ and b rank the types alike.

Normal Υ, regular roots (always if $\theta < 3.7(\beta+\delta_0)$). With redraws, (ii)'s claim on $V^{\female}$ and (iv)'s ranking condition held in every computed cell. Redraws off: Prop. 3.4.

Cheaper Contact, Less Partnership: An Example

Four types, sorting, everything visible: $\omega \in \{1, 2\}$, $\tau \in \{0.5, 0.75\}$, $g = b$, $f_0 = 1$; $\xi = 4$, $\eta = 2$, $\beta = 0.05$, $\delta_0 = 0.05$, $\theta = 1$, $\sigma_y = 0.5$.

γ^x (both sexes)	0.05	0.1	0.2	0.4	0.8	free
share partnered	0.900	0.923	**0.933**	0.931	0.920	0.905
m, bottom / top	0.39 / 0.52	0.54 / 0.67	0.74 / 0.86	0.99 / 1.11	1.32 / 1.42	4 / 4
$\delta(z, z)$, bottom / top	0.050 / 0.050	0.050 / 0.052	0.054 / 0.061	0.073 / 0.084	0.118 / 0.121	0.483 / 0.5

- Past $\gamma = 0.2$ each step raises the hazards proportionally more than the match rates: the share partnered falls.
- At $\gamma = 0.4$ the top's faster matching brings faster break-ups, and the bottom is partnered slightly more (0.932 against 0.930).

Sorting survives the refinement at every γ; the penalties do not move these numbers.

`verify_endog_delta_basepref.py`: the share partnered falls on some step in 224 of 384 cost paths with $\theta = 1$, $\sigma_y = 0.5$; the largest fall from a path's peak is 0.028, in this economy.

Two Indices Along the Cost of Contact

The economy of the previous slide; bottom ($z_{L,L}$) and top ($z_{H,H}$) men.

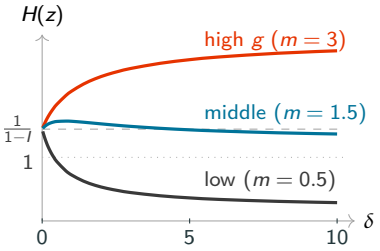
γ^x	0.05	0.1	0.2	0.4	0.8	
H_{time}, bottom / top	0.985 / 1.013	0.991 / 1.006	0.998 / 1.001	1.001 / 0.999	0.998 / 1.002	0.9
H_{flow}, bottom / top	0.983 / 1.015	0.972 / 1.023	0.929 / 1.051	0.914 / 1.050	0.969 / 1.003	1.1

- Costly contact: the top leads on both; the two indices nearly coincide while hazards are near δ_0.
- Contact free: everyone has $m = \xi$, and the gap is all in δ. The top leads on time partnered, the bottom on new matches.

Across the grid with $\theta \geq 0.1$, $\sigma_y = 0.5$: the top leads on time in every cell but 224, all at $\theta = 1$ and intermediate γ (the bottom leads in 128); the flow leader is the top at $\gamma = 0.05$ in 704 of 768 cells and the bottom with contact free in all 384.

Hegemony at Given Match Rates, Redraws Off

- Match rates held fixed; one common hazard δ, so the two indices coincide.
- As $\delta \to 0$ every non-celibate type's index tends to $(1-l)^{-1}$; faster dissolution raises the fastest type's index and lowers the slowest's (Prop. 3.5).
- With redraws, hazards differ by pair and move with match rates (E5): the indices separate.



Four types, a quarter of men each; one never matches: $l = \frac{1}{4}$.

The Screening Decision

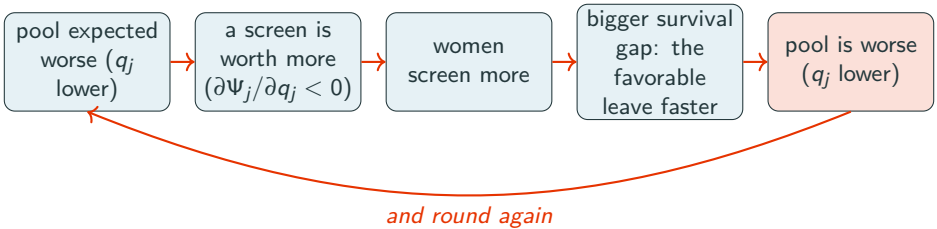
- Her surplus from the man: $\Delta_G > 0$ if he is favorable, $\Delta_B < 0$ if not. **Without a signal** she accepts iff $q_j \geq \widehat{q}_j \equiv |\Delta_B|/(\Delta_G + |\Delta_B|)$.
- **A signal** (correct w.p. σ_j) is worth

$$\Psi_j = \begin{cases} (1-q_j)\,\sigma_j\,|\Delta_B| - q_j\,(1-\sigma_j)\,\Delta_G & \text{if } q_j \geq \widehat{q}_j\text{: it weeds out the bad,} \\ q_j\,\sigma_j\,\Delta_G - (1-q_j)(1-\sigma_j)\,|\Delta_B| & \text{if } q_j < \widehat{q}_j\text{: it finds the good.} \end{cases} \tag{59}$$

- By (40), below $\widehat{q}_j$ the screen flips from a **filter** the bad slip through to a **barrier** they cross only by mistake.

Δ_G, Δ_B: surpluses at formation. Redraws off, stakes $\Delta_G - \Delta_B$ are $\Delta\alpha_j/(\beta+\delta_0)$ for her, $d_j/(\beta+\delta_0)$ for him, and only men with $f_0 - d_j < V^{\sigma}(z^{\sigma}) < f_0$ screen: men screen defensively, women acquisitively; with redraws each bound moves by the option value of a redraw. Exclusion is exact only if screening is exact ($\sigma_j = 1$) and, above $\widehat{q}_j$, exhaustive ($s_j = 1$).

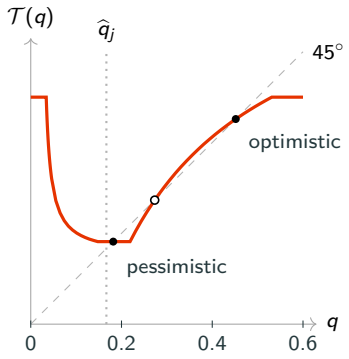
The Adverse-Selection Loop



- The loop runs while she **accepts unscreened men** ($q_j \geq \widehat{q}_j$): a **strategic complementarity**.
- Contact cannot select on a hidden attribute: the pool worsens through who *leaves*, not whom women contact. With redraws, also through who *stays*: she leaves the unfavorable sooner.

Several Equilibria: Expectations That Fulfill Themselves

- Pool and screening determine each other.
- **Expect a bad pool** $\Rightarrow$ screen hard $\Rightarrow$ the favorable leave fast $\Rightarrow$ a bad pool.
- **Expect a good pool** $\Rightarrow$ screen little $\Rightarrow$ all leave alike $\Rightarrow$ a good pool.
- $\mathcal{T}(q)$: the pool share a conjectured share q generates. It rises where unscreened men are accepted, so it can cross the 45° line **three times**.



Redraws off. Needs an interior screening margin and durable matches; a large δ restores uniqueness (Prop. A.1). Plotted at $\delta = 0.05$: a low-g man is partnered 75% vs. 94% of the time. Not recomputed with redraws.

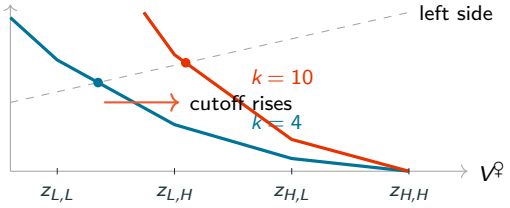
Where Women's Cutoff Comes From

- Her cutoff is her **search value** less the option value of a redraw: she accepts a man iff $g(z^{\male}) + \frac{\theta}{r^{\female}}\mathbb{E}[(y')^+] > V^{\female}(z^{\female})$. Written without an argument, $V^{\female}$ is the **lowest** search value in the sub-market.
- Her search value = meeting rate $\times$ expected surplus at formation of the men she accepts — which depends on the cutoff. So the cutoff is a **fixed point**.
- Simplest case (types learned on contact, men accept every woman, contact free, redraws off):

$$V^{\female} = k \sum_{z^{\male}} \pi(z^{\male}|S)\left[g(z^{\male}) - V^{\female}\right]^+, \qquad k \equiv \frac{\lambda^{\female}(\Theta(S))}{\beta + \delta_0}. \tag{60}$$

- The left side rises with $V^{\female}$, the right side falls: **one crossing**. Men with $g < V^{\female}$ add nothing to the right side: she rejects them. With redraws the surplus at formation replaces $[g - V^{\female}]^+/(\beta + \delta_0)$, and men just below $V^{\female}$ still add a little: probation.

What Moves the Cutoff



$g = (1.5, 2, 2.5, 3)$, $\pi = (.39, .29, .20, .12)$ (§4.3.1);

redraws off

- The cutoff is where the right side (solid) crosses the left (dashed): **1.67** at $k = 4$; only $z_{L,L}$ is rejected.
- Faster meeting or **cheaper contact** shifts the right side up: **2.05** at $k = 10$; $z_{L,H}$ falls below.
- More men at the bottom of the pool shift it down: the pool pushes back (Cor. A.1).

At a given pool, cheaper contact raises women's cutoffs: the **direct effect**.

Redraws off. With redraws (`dp_direct.py`, $\gamma^{♀}$ doubled at a fixed pool, men accepting the women alike): the lowest cutoff rose in every such sub-market at $\theta = 0.1$; at $\theta = 1$ it fell in up to 12%, every fall gone once men's search values are held.

On an App: Photos Visible, Income Hidden

τ-directed search: desirability τ is visible (photos); productivity ω is hidden (income unverified), so $\mathcal{O} = \{\tau\}$. The attractive feed holds $z_{H,H}$ *and* $z_{L,H}$.

Model	**On the app**
Sub-market, indexed by τ	the feeds you appear in, set by your photos and filters
Meeting	his profile appears in her feed
Contact (both must commit)	a mutual right swipe, then opening a chat
Screening on ω	the chat or first date: learning his job and income
Acceptance (both say yes)	deciding to form the relationship

Women do not form the sub-market: both sides choose one. Women's power is at contact, screening, and acceptance.

A Celibate's Path on the App

Take $z_{L,H}$, **attractive but unproductive**, under τ-directed search:

1. **Sub-market.** He is in the attractive feed with the $z_{H,H}$ men he looks like. No "attractive, low-income" feed can exist.
2. **Swipes.** Women see only his photo, so they swipe on him as often as on $z_{H,H}$.
3. **Date.** Her screen reveals ω_L. Her cutoff is set against the $z_{H,H}$ men she could meet next: if $g(z_{L,H})$ plus the option value of a redraw is below V^{φ}, she says no. Just above it, she says yes **on probation**, and leaves at her first poor redraw.
4. **Match only by mistake**, when her screen misses him: w.p. $1 - s_\omega \sigma_\omega$ while she accepts unscreened men, $s_\omega(1 - \sigma_\omega)$ once she rejects them.

He is not searching up: he searches at his **visible** level and is rejected for his **hidden** one.

With ω visible too, he would have his own sub-market, and no one would be excluded (Prop. 3.6′).

Computed: Turnover and Exclusion

Smallest g_{hi}/g_{lo} excluding outright, at $\delta_0 =$		0.005	0.02	0.05	0.158	0.5	2
$\theta = 0$: $R^*(\delta_0)$		12.11	4.64	3.20	2.73	3.20	6.13
she shares the	$\theta = 0.1$	3.79	3.45	3.14	2.91	3.33	6.20
higher man's type	$\theta = 0.5$	3.44	3.34	3.22	3.15	3.58	6.41
	$\theta = 2$	14.23	5.52	3.80	3.22	3.70	6.73
she shares the	$\theta = 0.1$	3.65	3.38	3.11	2.93	3.36	6.23
lower man's type	$\theta = 0.5$	3.23	3.21	3.20	3.25	3.74	6.57
	$\theta = 2$	3.74	3.52	3.32	3.27	3.90	7.11

- At $\theta = 2$, when she shares the higher man's type, redraws rarely end the accepted match, so the left arm returns: the threshold is finite at $\delta_0 = 0$ but large.
- When she shares the lower man's type, the higher man values her $f_0 - d$ and leaves her more often: the threshold falls where matches are durable and rises where δ_0 is large.

`dp_rstar.py`: $\beta = 0.05$, $\lambda = 1$, $f_0 = 1$, $d = 0.5$, $g_{lo} = 1$, $\sigma_y = 0.5$; $\theta = 0$ matches the formula to 10^{-14}; $\delta^* = (\beta\lambda/2)^{1/2} = 0.158$.

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